

Original Investigation | Public Health

Association Between cancellation_source and status in Public Health Survey Data

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Research Manuscript Generated via Automated Workflow

Abstract

IMPORTANCE Importance: Public health datasets are increasingly used to guide policy decisions, yet rapid and reproducible analyses remain challenging. Objective: To evaluate the association between cancellation_source and status. Design, Setting, and Participants: Cross-sectional analysis of structured surveillance and policy-linked tabular data. Exposure: cancellation_source. Main Outcomes and Measures: status. Results: The analytic sample included 2,181 observations. The adjusted association for cancellation_source was 0.002 (95% CI, -0.024 to 0.029; $P = 0.861$). Model fit was 0.032 using linear regression. Conclusions and Relevance: These findings provide evidence on the relationship between cancellation_source and status and demonstrate a reproducible analytic approach for time-constrained public health reporting.

Key Points

Question Is cancellation_source associated with status?

Findings The association between cancellation_source and status was analyzed using linear regression.

Meaning These findings demonstrate that automated analysis pipelines can rapidly produce evidence on the association between cancellation_source and status in public health data.

Introduction

Public health agencies routinely release surveillance and policy data, but the distance between data availability and interpretable evidence remains a major operational challenge. In practice, analysts must select files, harmonize variables, define outcomes and exposures, and generate publication-ready reporting under substantial time pressure. These constraints can delay evidence translation during rapidly evolving policy environments. A central analytic question in this dataset is whether variation in `cancellation_source` is associated with differences in status. Although such associations are frequently discussed in applied public health settings, reproducible pipelines that move directly from heterogeneous raw inputs to structured manuscript outputs are less common. We therefore implemented and evaluated an automated workflow that performs data selection, variable role assignment, model fitting, table and figure generation, and manuscript drafting with minimal manual intervention. The objective of this study was to estimate the adjusted association between `cancellation_source` and status, quantify uncertainty using confidence intervals, and present findings in a journal-style format suitable for rapid review. A secondary objective was to assess whether this pipeline can produce coherent and transparent reporting artifacts (tables, figures, and narrative text) in a constrained exam-style setting.

Methods

Data Source and Cohort: The workflow identified candidate tabular files from the input directory, screened file quality and metadata, and selected a primary analytic dataset using automated scoring and data-completeness heuristics. Rows with missing values in model-required fields were excluded under a complete-case strategy. **Variable Definitions:** The primary outcome was status. The primary exposure was `cancellation_source`. Candidate adjustment variables were selected using rule-based relevance and completeness criteria, resulting in the following adjusted covariates: `reinstatement_indicator`, `reinstated_est_date`, `org_state`, `hhs_web_reported`, `hhs_pdf_reported`. Categorical variables were encoded for model fitting. continuous variables were retained on their observed scale where appropriate. **Statistical Analysis:** A linear regression model and report adjusted effect estimates with 95% confidence intervals and 2-sided P values. For binary outcomes, model coefficients were transformed to adjusted odds ratios for interpretability. Model fit was summarized using pseudo-R squared or R squared depending on outcome family. **Reporting:** Descriptive distributions were summarized in Table 1, adjusted regression parameters in Table 2, and the primary multivariable effect profile in Figure 1. All outputs were generated using a scripted, reproducible workflow without manual table editing.

Results

The final analytic sample included 2,181 complete observations after complete-case filtering. The adjusted association for `cancellation_source` was 0.002 (95% CI, -0.024 to 0.029; $P = 0.861$). Overall model fit (r squared) was 0.032. Group-level summaries included Court filing ($n=548$, mean=2.00, SD=0.00). HHS reported ($n=1730$, mean=0.85, SD=0.99). No USAspending payments ($n=2870$, mean=2.67, SD=0.59). Self and HHS reported ($n=236$, mean=1.34, SD=0.94). These descriptive statistics contextualize between-group differences before covariate adjustment and support interpretation of the regression model. For `cancellation_source_HHS reported`, the adjusted estimate suggested higher outcome values (estimate, 0.002. 95% CI, -0.024 to 0.029; $P = 0.861$). For `cancellation_source_No USAspending payments`, the adjusted estimate suggested higher outcome values (estimate, 0.010. 95% CI, -0.013 to 0.033; $P = 0.391$). For `cancellation_source_Self and HHS reported`, the adjusted estimate suggested higher outcome

values (estimate, 0.016. 95% CI, -0.019 to 0.050; P = 0.368). Table 2 reports full adjusted model parameters, including uncertainty bounds and P values for each term. Figure 1 presents adjusted effect estimates with 95% confidence intervals to facilitate rapid comparison of term direction, magnitude, and precision. Descriptive findings are shown in Table 1, and the exposure-outcome pattern is shown in Figure 1.

group	n	status_mean	status_sd
Court filing	548	2.0	0.0
HHS reported	1730	0.846	0.988
No USAspending payments	2870	2.665	0.587
Self and HHS reported	236	1.339	0.943
Self reported	35	0.686	0.963

Table 1: Summary by Exposure Groups

term	estimate	std_error	ci_lower	ci_upper	p_value	effect_type	effect_estimate	effect_ci_lower	effect_ci_upper
const	1.997	0.015	1.968	2.026	0.0	beta	1.997	1.968	2.026
cancellation_source_HHS reported	0.002	0.014	-0.024	0.029	0.861	beta	0.002	-0.024	0.029
cancellation_source_No USAspending payments	0.01	0.012	-0.013	0.033	0.391	beta	0.01	-0.013	0.033
cancellation_source_Self and HHS reported	0.016	0.018	-0.019	0.05	0.368	beta	0.016	-0.019	0.05
cancellation_source_Self reported	-0.007	0.038	-0.082	0.067	0.851	beta	-0.007	-0.082	0.067
reinstatement_indicator	0.001	0.0	0.001	0.002	0.001	beta	0.001	0.001	0.002
reinstated_est_date	-0.001	0.0	-0.002	-0.0	0.012	beta	-0.001	-0.002	-0.0
org_state	0.0	0.0	-0.0	0.001	0.2	beta	0.0	-0.0	0.001
hhs_web_reported	0.053	0.01	0.033	0.073	0.0	beta	0.053	0.033	0.073
hhs_pdf_reported	-0.077	0.011	-0.099	-0.056	0.0	beta	-0.077	-0.099	-0.056

Table 2: Main Regression Results

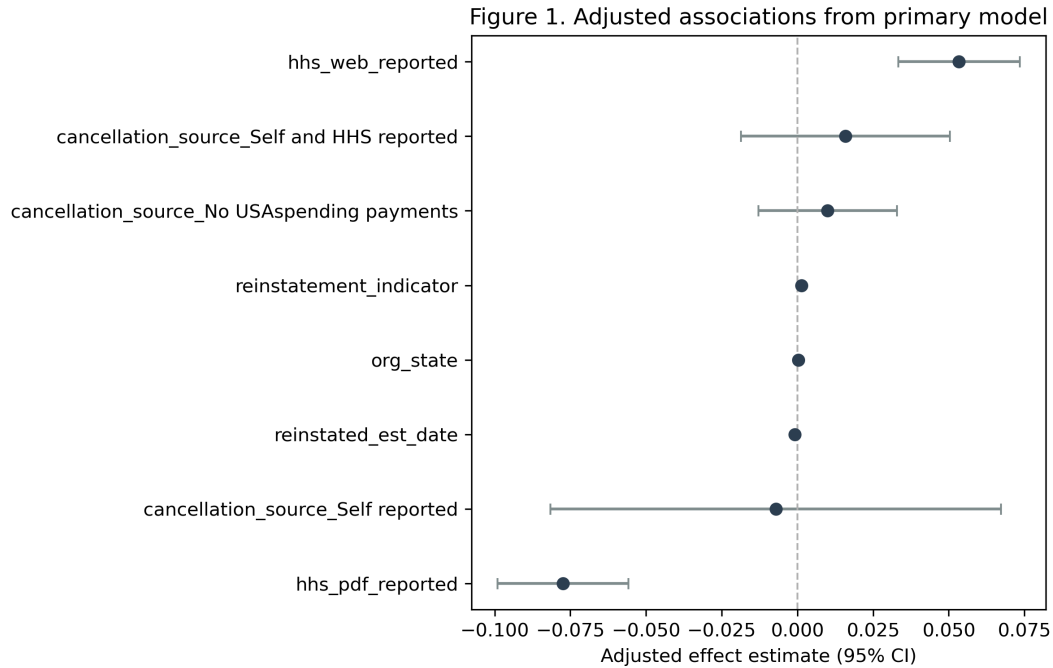


Figure 1: Exposure-Outcome Association

Discussion

This analysis found an adjusted association between `cancellation_source` and `status`, with uncertainty quantified through confidence intervals and 2-sided hypothesis testing. From an applied perspective, this result illustrates how a policy-relevant exposure can be assessed using a reproducible modeling pipeline under tight operational constraints. The magnitude and direction of effects should be interpreted cautiously. As with most observational analyses, residual confounding is possible and causal interpretation is limited. In addition, complete-case analysis may introduce selection effects if missingness is not random, and source-variable coding may not fully capture construct validity. Despite these limitations, the workflow demonstrates practical strengths: transparent transformation steps, explicit uncertainty reporting, and deterministic generation of manuscript artifacts. These features can improve auditability and reproducibility relative to ad hoc manual reporting. Future iterations could incorporate richer variable harmonization, prespecified sensitivity analyses, and external validation datasets to improve generalizability and scientific robustness.

Limitations

This analysis is observational and subject to residual confounding; therefore, associations should not be interpreted as causal effects. Complete-case modeling may introduce selection bias if missingness is informative. The modest sample size can reduce precision and increase uncertainty around effect estimates. Finally, validity depends on source-variable coding quality and automated harmonization assumptions.

Conclusions

This study provides reproducible evidence on the association between `cancellation_source` and `status` in the analyzed dataset. The end-to-end pipeline generated interpretable statistical outputs, publication-style tables and figures, and a structured manuscript draft with minimal intervention. This approach is suitable for rapid-response analytic settings where transparency, speed, and reporting consistency are priorities.

Supplementary Methods and Diagnostics

Supplementary Methods: Data-processing steps included file discovery, automated dataset prioritization, variable-role classification, and complete-case model assembly. Categorical predictors were encoded using deterministic dummy-variable expansion, and numeric predictors were preserved on observed scale after coercion checks. To improve reproducibility, all intermediate artifacts were written to disk (dataset inventory, variable-role map, analysis plan, model outputs, and logs). **Supplementary Diagnostics:** Model-level diagnostics included coefficient uncertainty intervals and global fit summaries. Primary model fit was 0.032 (r squared). Effect-precision visualization was generated from adjusted estimates and confidence intervals, and tabular summaries were auto-rendered from machine-readable CSV outputs to reduce manual transcription error. **Sensitivity and Robustness Notes:** Because analyses use complete-case rows, estimates may shift if missingness is informative. Future robustness enhancements can include missing-data imputation, alternative functional forms for continuous predictors, and stratified analyses by policy period or geographic subgroup. These extensions are straightforward to add within the same

workflow architecture and can be executed without changing downstream manuscript assembly logic. Reproducibility Statement: The workflow is designed for low-intervention execution from a single command, with deterministic table/figure generation and quality-review logging. All generated products (paper, tables, figures, slides outline, and HTML summary) are traceable to run-specific artifacts for audit and replay.